15.9 The Determinant as a Top Exterior Power

Chapter 7 built the determinant twice. First it proved that an alternating n-linear form on F^n is determined by its value at the standard basis (Theorem 7.4.2), which says *at most one* normalized such form exists; then it wrote the Leibniz sum down and checked the three rules (Theorem 7.4.3), which says *at least one* does. The number $\det \mathbf{A}$ of Definition 7.4.1 is that form's value. This section builds the determinant a third time, and the honest way to describe what it adds is to say up front what it does not: no new value, no faster computation, and no theorem Chapter 7 was missing. What it adds is an **object**. The exterior powers of the previous section turn $\det T$ from a number attached to a matrix into the operator that T induces on a one-dimensional space, and from that description multiplicativity stops being a computation and becomes the statement that composing two induced maps induces the composite.

Throughout, F is a field, V and W are finite-dimensional vector spaces over F, and $\Lambda^k V$, the wedge $v_1 \wedge \cdots \wedge v_k$, and the universal property for alternating maps are as in Definition 15.8.2. We use freely that the wedges $v_{i_1} \wedge \cdots \wedge v_{i_k}$ over strictly increasing index tuples form a basis of $\Lambda^k V$, so that $\dim \Lambda^k V = \binom{n}{k}$ when $\dim V = n$ (Theorem 15.8.5), and that a wedge of vectors is nonzero exactly when the list is linearly independent (Theorem 15.8.7).

15.9.1 Wedging with a fixed vector

Two facts from the previous section will be used on every page, and one small tool is still missing.

The first fact is the **reordering rule**: the wedge map $(v_1, \ldots, v_k) \mapsto v_1 \wedge \cdots \wedge v_k$ is alternating (Definition 15.8.2), and an alternating multilinear map satisfies

$$f(v_{\sigma(1)}, \ldots, v_{\sigma(k)}) = \operatorname{sgn}(\sigma)\, f(v_1, \ldots, v_k)$$

for every $\sigma \in S_k$, by Lemma 15.8.1 (b). Reordering a wedge costs the sign of the permutation, and nothing else. The second is that $\Lambda^n V$ is one-dimensional when $\dim V = n \geq 1$ (Corollary 15.8.6); that is the whole engine of this section.

The missing tool is the ability to wedge a fixed vector onto a whole exterior power, not just onto a list of vectors. Section 10 builds the product on all of ΛV; here one vector at a time is enough.

Lemma 15.9.1. Wedging with a Fixed Vector

Let $v \in V$ and $k \geq 1$. There is a **unique** linear map

$$\varepsilon_v \colon \Lambda^k V \to \Lambda^{k+1} V, \qquad \varepsilon_v(w_1 \wedge \cdots \wedge w_k) = v \wedge w_1 \wedge \cdots \wedge w_k.$$

Proof. The map $V^k \to \Lambda^{k+1} V$ sending $(w_1, \ldots, w_k)$ to $v \wedge w_1 \wedge \cdots \wedge w_k$ is k-linear, because the wedge of $k+1$ vectors is linear in each slot and the first slot is frozen at v; and it is alternating, because if $w_p = w_r$ with $p \neq r$ then the wedge of $k+1$ vectors has two equal arguments and is 0. By the universal property in Definition 15.8.2 it factors through $\Lambda^k V$, giving a linear ε_v with the stated values. Uniqueness holds because the wedges $w_1 \wedge \cdots \wedge w_k$ span $\Lambda^k V$ (Theorem 15.8.5), and a linear map is determined on a spanning set.

■

15.9.2 Induced maps on exterior powers

A linear map should act on everything built from its domain. A map $T\colon V \to W$ sends a list $(v_1, \dots, v_k)$ to the list $(Tv_1, \dots, Tv_k)$, so it ought to send the wedge $v_1 \wedge \cdots \wedge v_k$ to $Tv_1 \wedge \cdots \wedge Tv_k$. The only question is whether that rule defines a map at all: an element of $\Lambda^k V$ can be written as a wedge in many ways, and an assignment made on wedges must be checked. The universal property answers it in two lines, and that is the point of having a universal property.

The k-th exterior power of a map wedges its values together.

Definition 15.9.1. Exterior Power of a Linear Map

Let $T\colon V \to W$ be linear and let $k \geq 1$. The **k-th exterior power** of T is the unique linear map

$$\Lambda^k T\colon \Lambda^k V \to \Lambda^k W$$

satisfying $\Lambda^k T(v_1 \wedge \cdots \wedge v_k) = Tv_1 \wedge \cdots \wedge Tv_k$ for all $v_1, \dots, v_k \in V$.

Well-definedness. The map $V^k \to \Lambda^k W$ sending $(v_1, \dots, v_k)$ to $Tv_1 \wedge \cdots \wedge Tv_k$ is k-linear, since T is linear and the wedge in W is linear in each slot; and it is alternating, since $v_p = v_r$ forces $Tv_p = Tv_r$ and a wedge with two equal arguments is 0. So Definition 15.8.2 produces exactly one linear map on $\Lambda^k V$ with the stated values, and the wedges span $\Lambda^k V$ (Theorem 15.8.5), so no other linear map has them.

Examples.

- $k = 1$. $\Lambda^1 V = V$ and $\Lambda^1 T = T$: the definition says nothing new in degree one.
- **The identity.** $\Lambda^k(\mathrm{id}_V) = \mathrm{id}$, since both sides send $v_1 \wedge \cdots \wedge v_k$ to itself.
- **A scalar.** For $T = c\,\mathrm{id}_V$, $\Lambda^k T(v_1 \wedge \cdots \wedge v_k) = c^k\, v_1 \wedge \cdots \wedge v_k$, so $\Lambda^k T = c^k\,\mathrm{id}$. One factor of c comes out of each of the k slots.
- **A degenerate case.** If $k > \dim V$ then $\Lambda^k V = \{0\}$ (Theorem 15.8.5) and $\Lambda^k T$ is the zero map between zero spaces. Worth recording so that statements below need no size caveat.

Nonexample by minimal change. Replace "alternating" by nothing: the rule $v_1 \otimes \cdots \otimes v_k \mapsto Tv_1 \wedge \cdots \wedge Tv_k$ on $V^{\otimes k}$ is perfectly well defined, but it is **not** the map above, because $\Lambda^k V$ is a quotient of $V^{\otimes k}$ and not a subspace of it. What makes Definition 15.9.1 legal is precisely that the rule kills the relations that were divided out.

The reason for having the notation is the next sentence.

Theorem 15.9.2. Exterior Powers Respect Composition

Let $T\colon U \to V$ and $S\colon V \to W$ be linear, and let $k \geq 1$. Then

$$\Lambda^k(ST) = (\Lambda^k S)(\Lambda^k T), \qquad \Lambda^k(\mathrm{id}_V) = \mathrm{id}_{\Lambda^k V}.$$

Proof. Both $\Lambda^k(ST)$ and $(\Lambda^k S)(\Lambda^k T)$ are linear maps $\Lambda^k U \to \Lambda^k W$, so it suffices to compare them on the wedges, which span $\Lambda^k U$ (Theorem 15.8.5). By Definition 15.9.1 applied three times,

$$(\Lambda^k S)(\Lambda^k T)(u_1 \wedge \cdots \wedge u_k) = STu_1 \wedge \cdots \wedge STu_k = \Lambda^k(ST)(u_1 \wedge \cdots \wedge u_k).$$

The second identity was checked above. This proves the theorem.

■

Example. The second exterior power of a 3×3 matrix

Let $T\colon \mathbb{Q}^3 \to \mathbb{Q}^3$ have matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & 1 \\ 5 & 2 & 1 \end{pmatrix}$$

in the standard basis. Find the matrix of $\Lambda^2 T$ in the basis $(\mathbf{e}_1 \wedge \mathbf{e}_2,\ \mathbf{e}_1 \wedge \mathbf{e}_3,\ \mathbf{e}_2 \wedge \mathbf{e}_3)$ of $\Lambda^2 \mathbb{Q}^3$, and compute its determinant.

Solution. By Definition 15.9.1, $\Lambda^2 T(\mathbf{e}_i \wedge \mathbf{e}_j) = \mathbf{a}_i \wedge \mathbf{a}_j$, where $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are the columns of $\mathbf{A}$. Expand one column:

$$\begin{aligned} \mathbf{a}_1 \wedge \mathbf{a}_2 &= (2\mathbf{e}_1 + 5\mathbf{e}_3) \wedge (\mathbf{e}_1 + 4\mathbf{e}_2 + 2\mathbf{e}_3) \\ &= 8\,\mathbf{e}_1 \wedge \mathbf{e}_2 + 4\,\mathbf{e}_1 \wedge \mathbf{e}_3 + 5\,\mathbf{e}_3 \wedge \mathbf{e}_1 + 20\,\mathbf{e}_3 \wedge \mathbf{e}_2 \\ &= 8\,\mathbf{e}_1 \wedge \mathbf{e}_2 - \mathbf{e}_1 \wedge \mathbf{e}_3 - 20\,\mathbf{e}_2 \wedge \mathbf{e}_3, \end{aligned}$$

where the terms $2\,\mathbf{e}_1 \wedge \mathbf{e}_1$ and $10\,\mathbf{e}_3 \wedge \mathbf{e}_3$ were dropped in the second line because a wedge with a repeated vector is 0, and the third line used $\mathbf{e}_3 \wedge \mathbf{e}_1 = -\mathbf{e}_1 \wedge \mathbf{e}_3$ and $\mathbf{e}_3 \wedge \mathbf{e}_2 = -\mathbf{e}_2 \wedge \mathbf{e}_3$ (Lemma 15.8.1 (b)). The coefficients $8, -1, -20$ are the three 2×2 minors of the first two columns of $\mathbf{A}$, on rows $\{1,2\}, \{1,3\}, \{2,3\}$ — a pattern the last part of this section explains. Doing the same for the other two column pairs gives

$$[\Lambda^2 T] = \begin{pmatrix} 8 & 2 & -11 \\ -1 & -13 & -5 \\ -20 & -5 & 2 \end{pmatrix},$$

whose determinant is 2601. Since $\det \mathbf{A} = -51$, this is $(\det \mathbf{A})^2$.

Quick check. In the example, why is $\Lambda^2 T$ **not** multiplication by a scalar, while $\Lambda^3 T$ is?

Solution. $\dim \Lambda^2 \mathbb{Q}^3 = \binom{3}{2} = 3$ by Theorem 15.8.5, and the displayed matrix is not a scalar multiple of $\mathbf{I}_3$, so $\Lambda^2 T$ is an operator on a three-dimensional space with no reason to be scalar. But $\dim \Lambda^3 \mathbb{Q}^3 = \binom{3}{3} = 1$, and every linear operator on a one-dimensional space is multiplication by a scalar. Being a line is what makes the top power special.

15.9.3 The top exterior power

That last observation is the whole theorem. When k equals $\dim V$, the space $\Lambda^k V$ is a line, an operator on a line is a scalar, and the scalar can be read off by feeding in a basis.

Theorem 15.9.3. The Determinant Is the Top Exterior Power

Let V be a vector space over F with $\dim V = n \geq 1$, and let $T \in \mathcal{L}(V)$. Then $\dim \Lambda^n V = 1$, and $\Lambda^n T$ is multiplication by $\det T$:

$$\Lambda^n T(\omega) = (\det T)\,\omega \qquad \text{for every } \omega \in \Lambda^n V.$$

Equivalently, for all $v_1, \ldots, v_n \in V$,

$$Tv_1 \wedge \cdots \wedge Tv_n = (\det T)\, v_1 \wedge \cdots \wedge v_n. \tag{15.9.1}$$

Idea. Three moves. ① $\Lambda^n V$ is one-dimensional, so $\Lambda^n T$ is multiplication by one unknown scalar c. ② To find c, evaluate on a basis wedge $v_1 \wedge \cdots \wedge v_n$ and expand each Tv_j in the basis. ③ The expansion produces one term for every function from slots to basis indices; the noninjective ones die because a wedge with a repeat is zero, and what survives is a sum over permutations with signs — the Leibniz formula, arriving on its own.

Proof. Fix a basis $\mathcal{B} = (v_1, \ldots, v_n)$ of V. By Corollary 15.8.6, $\dim \Lambda^n V = 1$ with basis the single wedge $\omega_0 = v_1 \wedge \cdots \wedge v_n$. Since $\Lambda^n T$ is a linear operator on a one-dimensional space, there is a unique $c \in F$ with $\Lambda^n T(\omega_0) = c\,\omega_0$, and then $\Lambda^n T(a\omega_0) = ac\,\omega_0$ for every $a \in F$, so $\Lambda^n T = c$ id on all of $\Lambda^n V$.

It remains to show $c = \det T$. Let $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}}$ have entries a_{ij}, so that $Tv_j = \sum_{i=1}^{n} a_{ij} v_i$ (Definition 4.1.1). The wedge of n vectors is linear in each slot, so expanding slot by slot gives one term for each function $\varphi\colon \{1, \ldots, n\} \to \{1, \ldots, n\}$:

$$Tv_1 \wedge \cdots \wedge Tv_n = \sum_{\varphi} a_{\varphi(1)1} \cdots a_{\varphi(n)n}\; v_{\varphi(1)} \wedge \cdots \wedge v_{\varphi(n)}.$$

If φ is not injective, two slots hold the same vector and the wedge is 0. A noninjective φ is possible only because the domain and codomain are finite of the same size, so the surviving φ are exactly the bijections, that is, the permutations $\sigma \in S_n$. For those, Lemma 15.8.1 (b) gives $v_{\sigma(1)} \wedge \cdots \wedge v_{\sigma(n)} = \operatorname{sgn}(\sigma)\,\omega_0$. Therefore

$$\Lambda^n T(\omega_0) = \Big(\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a_{\sigma(1)1} \cdots a_{\sigma(n)n} \Big) \omega_0,$$

and the bracket is $\det \mathbf{A}$ by Definition 7.4.1, which is $\det T$ by Definition 7.5.1. So $c = \det T$.

For the last display, note that both sides of Eq. 15.9.1 are n-linear in $(v_1, \ldots, v_n)$ and agree when the list is the basis $\mathcal{B}$; but in fact no extra argument is needed, since $v_1 \wedge \cdots \wedge v_n \in \Lambda^n V$ for **any** list, and the first display applies to it. This proves the theorem. ∎

Warning. **Only the top power is a line.** For $1 \leq k < n$ the operator $\Lambda^k T$ is not multiplication by a scalar in general: Example (The second exterior power of a 3×3 matrix)

exhibits a $\Lambda^2 T$ with three different diagonal entries. The determinant appears because $\binom{n}{n} = 1$, not because exterior powers are scalars.

Warning. **A map between different spaces still has no determinant.** If $T\colon V \to W$ is linear with $\dim V = \dim W = n$, then $\Lambda^n T$ is a linear map between two **different** one-dimensional spaces, and there is no scalar attached to it until bases of $\Lambda^n V$ and $\Lambda^n W$ are chosen. Changing the basis of $\Lambda^n W$ alone rescales the number arbitrarily. This is the same obstruction Chapter 7 recorded after Definition 7.5.1, now visible as the failure of a line to have a canonical generator.

Quick check. Let T be the operator on F^3 with $T\mathbf{e}_1 = \mathbf{e}_2$, $T\mathbf{e}_2 = \mathbf{e}_3$, and $T\mathbf{e}_3 = \mathbf{e}_1$. Read $\det T$ off Eq. 15.9.1, without writing down a matrix.

Solution. $T\mathbf{e}_1 \wedge T\mathbf{e}_2 \wedge T\mathbf{e}_3 = \mathbf{e}_2 \wedge \mathbf{e}_3 \wedge \mathbf{e}_1$. The permutation carrying $(1,2,3)$ to $(2,3,1)$ is a 3-cycle, of sign $+1$ (Proposition 7.3.4), so by Lemma 15.8.1 (b) this equals $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$. Comparing with Eq. 15.9.1 on the nonzero element $\mathbf{e}_1 \wedge \mathbf{e}_2 \wedge \mathbf{e}_3$ gives $\det T = 1$.

15.9.4 What the third construction adds

It is worth saying plainly, because the temptation to oversell a slick construction is real.

It adds no new value. The scalar produced in the proof is the Leibniz sum, arrived at by the same expansion that proved Theorem 7.4.2 in Chapter 7: expand multilinearly, discard the terms with a repeated basis vector, sort the survivors, and collect signs. Anyone who has read Chapter 7 §04 has read that computation. The exterior power did not compute the determinant; it stored the same computation in a different place.

It adds no faster method. Nothing here helps evaluate a determinant. Elimination remains the way to do that (Example (A 4×4 Determinant by Elimination)).

What it does add is that det **is a map, not just a number.** Chapter 7 defined $\det \mathbf{A}$ by a formula in the entries and then had to *prove* that operators inherit it: the formula depends on a basis, so Corollary 7.5.7 was needed before Definition 7.5.1 could be written down. Here $\Lambda^n T$ is defined with no basis in sight, and the scalar exists because $\Lambda^n V$ is a line. Basis-independence is not a theorem about the determinant; it is the absence of any choice to be independent of. One honest qualification: the route is not self-contained. Theorem 15.8.5, which is what makes $\Lambda^n V$ a line in the first place, proves independence with a $k \times k$ determinant, and the identification of the scalar with $\det T$ goes through Definition 7.5.1. Chapter 7 is not being replaced here; it is being re-read.

And from that, multiplicativity is free.

Corollary 15.9.4. Multiplicativity and Invertibility, from Functoriality

Let $\dim V = n \geq 1$ and $S, T \in \mathcal{L}(V)$.

(a) $\det(ST) = \det S \det T$, and $\det \mathrm{id}_V = 1$.
(b) T is invertible if and only if $\det T \neq 0$.

Proof.

(a) By Theorem 15.9.2, $\Lambda^n(ST) = (\Lambda^n S)(\Lambda^n T)$. By Theorem 15.9.3 all three maps are multiplications by scalars on the line $\Lambda^n V$, namely by $\det(ST)$, $\det S$, and $\det T$; and multiplication by $\det T$ followed by multiplication by $\det S$ is multiplication by $\det S \det T$. Two scalar multiplications on a nonzero space are equal only if the scalars are, so $\det(ST) = \det S \det T$. Likewise $\Lambda^n(\mathrm{id}_V) = \mathrm{id}$, so $\det \mathrm{id}_V = 1$.

(b) ($\Rightarrow$) If T is invertible, apply (a) to $TT^{-1} = \mathrm{id}_V$: then $\det T \cdot \det(T^{-1}) = 1$, so $\det T \neq 0$. ($\Leftarrow$) Suppose T is not invertible. Then T is not injective, so $\ker T \neq \{0\}$; choose $v_1 \in \ker T$ with $v_1 \neq 0$ and extend to a basis $(v_1, \dots, v_n)$ of V (Theorem 1.6.5). Since $Tv_1 = 0$ and the wedge is linear in its first slot, $Tv_1 \wedge \cdots \wedge Tv_n = 0$. By Eq. 15.9.1 this is $(\det T)\, v_1 \wedge \cdots \wedge v_n$, and the wedge of a basis is nonzero (Theorem 15.8.7), so $\det T = 0$.

■

Part (b) is the promise Theorem 15.8.7 made: the test "is this list independent?" and the test "is this determinant nonzero?" are the same test, once the determinant is read as an induced map.

Compare the work this replaces. Chapter 7 proved Theorem 7.5.4 by fixing $\mathbf{A}$, checking that $\mathbf{B} \mapsto \det(\mathbf{AB})$ is alternating and n-linear in the columns of $\mathbf{B}$, and applying uniqueness. That proof is short and correct; what the present one shows is that the content is not really about determinants at all. Multiplicativity is the statement that Λ^n turns composition into composition — and Λ^n does that for the same reason that Λ^k does it for every k, where no scalar is available and no multiplicativity statement can even be phrased.

Remark. What is **not** improved: Theorem 7.4.5. The identity $\det \mathbf{A}^\top = \det \mathbf{A}$ is a statement about rows versus columns, and Λ^n sees only the map T, which has no rows. Chapter 7's proof, reindexing the Leibniz sum by σ^{-1}, remains the natural one.

15.9.5 The coordinates of a wedge

The example above produced the minors of a matrix as the coefficients of a wedge, apparently by accident. It was not an accident, and the general statement is the bridge between this chapter and Chapter 7 §07.

Fix $n \geq 1$, write $[n] = \{1, \dots, n\}$ as Chapter 7 §07 does, and recall the notation of Definition 7.7.1: for $\mathbf{A} \in M_{n\times k}(F)$ and a set $I = \{i_1 < \cdots < i_k\} \subseteq [n]$ of size k, the submatrix $\mathbf{A}_{I,[k]} \in M_k(F)$ keeps the rows in I and all k columns, and its determinant is a $k \times k$ minor of $\mathbf{A}$. Write

$$e_I := \mathbf{e}_{i_1} \wedge \mathbf{e}_{i_2} \wedge \cdots \wedge \mathbf{e}_{i_k} \in \Lambda^k F^n,$$

so that the e_I, over all k-element $I \subseteq [n]$, form the basis of Theorem 15.8.5.

Theorem 15.9.5. The Coordinates of a Wedge Are the Minors

Let $1 \le k \le n$, let $\mathbf{v}_1, \dots, \mathbf{v}_k \in F^n$, and let $\mathbf{A} \in M_{n\times k}(F)$ be the matrix with these vectors as its columns. Then

$$\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \cdots \wedge \mathbf{v}_k = \sum_I (\det \mathbf{A}_{I,[k]})\, e_I,$$

the sum running over all subsets $I \subseteq [n]$ with $|I| = k$. That is, the coordinate of $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$ at e_I is the $k \times k$ minor of $\mathbf{A}$ on the rows I.

Idea. The same expansion as in Theorem 15.9.3, but now the index functions run into a bigger set. An injective $\varphi\colon [k] \to [n]$ is a choice of image set I followed by an ordering of it, so the terms group into one bunch per I, and each bunch is a signed sum over S_k: a determinant of size k.

Proof. Write $\mathbf{A} = (a_{ij})$, so $\mathbf{v}_j = \sum_{i=1}^n a_{ij}\mathbf{e}_i$. Expanding the wedge in each of its k slots gives one term per function $\varphi\colon [k] \to [n]$:

$$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k = \sum_{\varphi} a_{\varphi(1)1} \cdots a_{\varphi(k)k}\, \mathbf{e}_{\varphi(1)} \wedge \cdots \wedge \mathbf{e}_{\varphi(k)}.$$

A noninjective φ repeats a basis vector, so its wedge is 0 and its term drops out. An injective φ is determined by its image $I = \{i_1 < \cdots < i_k\}$ together with the permutation $\sigma \in S_k$ defined by $\varphi(j) = i_{\sigma(j)}$; conversely every pair (I, σ) arises from exactly one injective φ. For such a φ, Lemma 15.8.1 (b) gives

$$\mathbf{e}_{\varphi(1)} \wedge \cdots \wedge \mathbf{e}_{\varphi(k)} = \mathbf{e}_{i_{\sigma(1)}} \wedge \cdots \wedge \mathbf{e}_{i_{\sigma(k)}} = \operatorname{sgn}(\sigma)\, e_I.$$

Grouping the terms by I therefore gives

$$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k = \sum_I \Big(\sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)\, a_{i_{\sigma(1)}1} \cdots a_{i_{\sigma(k)}k} \Big) e_I.$$

By Definition 7.7.1 the (p,q)-entry of $\mathbf{A}_{I,[k]}$ is $a_{i_p q}$, so the inner sum is exactly the Leibniz sum of Definition 7.4.1 for the $k\times k$ matrix $\mathbf{A}_{I,[k]}$, that is, $\det \mathbf{A}_{I,[k]}$. This proves the theorem. ∎

Example. A wedge of two vectors in four-dimensional space

In $\mathbb{Q}^4$, let $\mathbf{v}_1 = (1, 0, 2, 1)$ and $\mathbf{v}_2 = (0, 1, 1, 3)$. Write $\mathbf{v}_1 \wedge \mathbf{v}_2$ in the basis of $\Lambda^2\mathbb{Q}^4$.

Solution. Let $\mathbf{A} \in M_{4\times 2}(\mathbb{Q})$ have these two vectors as its columns, so that $\mathbf{A}_{\{1,2\},[2]} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and the other five 2×2 submatrices read off the same way. Their determinants are

$$\det \mathbf{A}_{\{1,2\}} = 1, \quad \det \mathbf{A}_{\{1,3\}} = 1, \quad \det \mathbf{A}_{\{1,4\}} = 3,$$
$$\det \mathbf{A}_{\{2,3\}} = -2, \quad \det \mathbf{A}_{\{2,4\}} = -1, \quad \det \mathbf{A}_{\{3,4\}} = 5,$$

abbreviating $\mathbf{A}_{I,[2]}$ to $\mathbf{A}_I$. By Theorem 15.9.5,

$$\begin{aligned}\mathbf{v}_1 \wedge \mathbf{v}_2 = e_{12} + e_{13} + 3\,e_{14} \\ - 2\,e_{23} - e_{24} + 5\,e_{34}.\end{aligned}$$

As a check, some coordinate is nonzero, so $\mathbf{v}_1 \wedge \mathbf{v}_2 \neq 0$, so $(\mathbf{v}_1, \mathbf{v}_2)$ is linearly independent by Theorem 15.8.7 — which is visible directly, since neither vector is a multiple of the other.

That check generalizes at once, and what comes out is a theorem of Chapter 7.

Corollary 15.9.6. Independence via Minors

Let $\mathbf{v}_1, \ldots, \mathbf{v}_k \in F^n$ and let $\mathbf{A} \in M_{n\times k}(F)$ have these vectors as its columns. Then $(\mathbf{v}_1, \ldots, \mathbf{v}_k)$ is linearly independent if and only if some $k \times k$ minor of $\mathbf{A}$ is nonzero.

Proof. By Theorem 15.8.7 the list is independent exactly when $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k \neq 0$. By Theorem 15.9.5 the coordinates of that wedge in the basis (e_I) of Theorem 15.8.5 are the $k \times k$ minors of $\mathbf{A}$, and a vector is 0 exactly when all of its coordinates in a basis vanish. (For $k > n$ both sides fail: $\Lambda^k F^n = \{0\}$ and $\mathbf{A}$ has no $k \times k$ submatrix.)

■

This is Theorem 7.7.1 in the case of full column rank, and the general case follows from it: $\operatorname{rank} \mathbf{A}$ is the largest number of independent columns, hence the largest k for which some k columns carry a nonzero $k \times k$ minor. Chapter 7 proved it with two careful steps involving pivot columns and row rank. The wedge proves it by naming the coordinates.

15.9.6 Laplace expansion

Cofactor expansion also comes out of the coordinate theorem, once the wedge of $n-1$ vectors is available. The route is: pull the chosen column out of the wedge, recognize what is left as an element of $\Lambda^{n-1}F^n$ whose coordinates are the $(n-1)\times(n-1)$ minors, and wedge $\mathbf{e}_i$ back on.

Corollary 15.9.7. Cofactor Expansion, from a Wedge

Let $n \geq 2$, let $\mathbf{A} \in M_n(F)$ with minors M_{ij} as in Definition 7.6.1, and fix $j \in [n]$. Then

$$\det \mathbf{A} = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} M_{ij}.$$

Idea. Write $\det \mathbf{A}$ as the wedge of the columns. Linearity in the j-th slot splits it into n pieces, one per i, with $\mathbf{e}_i$ sitting in that slot. Move $\mathbf{e}_i$ to the front at a cost of $(-1)^{j-1}$; the remaining $n-1$ columns form a wedge whose coordinates are the minors of the matrix with column j deleted, and wedging $\mathbf{e}_i$ on kills every coordinate except the one whose index set misses i.

Proof. Let $\mathbf{a}_1, \dots, \mathbf{a}_n$ be the columns of $\mathbf{A}$ and $\omega_0 = \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_n$. By Theorem 15.9.3 applied to the operator $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$, whose matrix in the standard basis is $\mathbf{A}$,

$$\mathbf{a}_1 \wedge \cdots \wedge \mathbf{a}_n = (\det \mathbf{A})\, \omega_0.$$

Substituting $\mathbf{a}_j = \sum_{i=1}^{n} a_{ij} \mathbf{e}_i$ and using linearity in slot j,

$$(\det \mathbf{A})\, \omega_0 = \sum_{i=1}^{n} a_{ij}\, \mathbf{a}_1 \wedge \cdots \wedge \underset{\text{slot } j}{\mathbf{e}_i} \wedge \cdots \wedge \mathbf{a}_n.$$

Fix i. Moving $\mathbf{e}_i$ from slot j to the front means $j-1$ exchanges of neighbors, so by Lemma 15.8.1 (b) the i-th wedge above equals $(-1)^{j-1}\, \varepsilon_{\mathbf{e}_i}(\eta)$, where $\eta = \mathbf{a}_1 \wedge \cdots \wedge \widehat{\mathbf{a}_j} \wedge \cdots \wedge \mathbf{a}_n \in \Lambda^{n-1} F^n$ omits the j-th column, and $\varepsilon_{\mathbf{e}_i}$ is the map of Lemma 15.9.1.

Let $\mathbf{B} \in M_{n \times (n-1)}(F)$ be $\mathbf{A}$ with column j deleted. By Theorem 15.9.5,

$$\eta = \sum_{l=1}^{n} (\det \mathbf{B}_{[n]\setminus\{l\},\,[n-1]})\, e_{[n]\setminus\{l\}},$$

since the $(n-1)$-element subsets of $[n]$ are exactly the complements of single elements. Deleting row l from $\mathbf{B}$ deletes row l and column j from $\mathbf{A}$, so $\det \mathbf{B}_{[n]\setminus\{l\},[n-1]} = M_{lj}$ by Definition 7.6.1. Now apply $\varepsilon_{\mathbf{e}_i}$. For $l \neq i$ the index set $[n] \setminus \{l\}$ contains i, so $\mathbf{e}_i \wedge e_{[n]\setminus\{l\}}$ repeats $\mathbf{e}_i$ and vanishes; for $l = i$, moving $\mathbf{e}_i$ into its place costs $i-1$ exchanges, so $\mathbf{e}_i \wedge e_{[n]\setminus\{i\}} = (-1)^{i-1}\omega_0$ by Lemma 15.8.1 (b). Hence $\varepsilon_{\mathbf{e}_i}(\eta) = (-1)^{i-1} M_{ij}\, \omega_0$, and

$$(\det \mathbf{A})\, \omega_0 = \sum_{i=1}^{n} a_{ij} (-1)^{j-1} (-1)^{i-1} M_{ij}\, \omega_0.$$

Since $(-1)^{i+j-2} = (-1)^{i+j}$ and $\omega_0 \neq 0$ (Theorem 15.8.7), comparing coefficients gives the stated formula.

■

This is Theorem 7.6.1 (a). The row version follows from it by Theorem 7.4.5, exactly as in Chapter 7. The gain is again structural rather than practical: the sign $(-1)^{i+j}$, which in Chapter 7 arrived from a careful count of row and column moves, is here the price of two reversals, $j-1$ in one direction and $i-1$ in the other.

15.9.7 Exercises

15.9.7.1 A. Check your understanding

Exercise. A1

(a) State the defining property of $\Lambda^k T$, and say why a rule given on wedges needs an argument before it defines a map.

(b) State Theorem 15.9.3.

(c) Name one thing this section's construction of the determinant adds to Chapter 7's, and one thing it does not.

(d) True or false: for every $T \in \mathcal{L}(V)$ and every k, the map $\Lambda^k T$ is multiplication by a

scalar. Justify your answer.

(e) What are the coordinates of $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_k$ in $\Lambda^k F^n$, in the basis (e_I)?

Solution.

(a) $\Lambda^k T$ is the unique linear map $\Lambda^k V \to \Lambda^k W$ with $\Lambda^k T(v_1 \wedge \cdots \wedge v_k) = Tv_1 \wedge \cdots \wedge Tv_k$ (Definition 15.9.1). An element of $\Lambda^k V$ has many expressions as a sum of wedges, so a rule stated on wedges might assign two different values to one element; the universal property of Definition 15.8.2 removes the danger by producing the map from an alternating multilinear map upstairs.

(b) If $\dim V = n \geq 1$ and $T \in \mathcal{L}(V)$, then $\dim \Lambda^n V = 1$ and $\Lambda^n T(\omega) = (\det T)\omega$ for every $\omega \in \Lambda^n V$.

(c) It adds that the determinant is the induced map on a line, so that no basis and no similarity-invariance argument is needed to define $\det T$, and multiplicativity becomes Theorem 15.9.2. It does not add a new value, a new method of computation, or any improvement to $\det \mathbf{A}^\top = \det \mathbf{A}$.

(d) False. It is true when $k = n = \dim V$, and when $k > n$ for the trivial reason that $\Lambda^k V = \{0\}$. For $1 \leq k < n$ it usually fails: see Example (The second exterior power of a 3×3 matrix), where $\Lambda^2 T$ has matrix with rows $(8, 2, -11)$, $(-1, -13, -5)$, $(-20, -5, 2)$, not a scalar matrix.

(e) The $k \times k$ minors of the matrix whose columns are $\mathbf{v}_1, \ldots, \mathbf{v}_k$: the coordinate at e_I is $\det \mathbf{A}_{I,[k]}$ (Theorem 15.9.5).

15.9.7.2 B. Practice

Exercise. B1: A second exterior power

Let $T\colon \mathbb{Q}^3 \to \mathbb{Q}^3$ have standard matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}.$$

Find the matrix of $\Lambda^2 T$ in the basis $(\mathbf{e}_1 \wedge \mathbf{e}_2, \mathbf{e}_1 \wedge \mathbf{e}_3, \mathbf{e}_2 \wedge \mathbf{e}_3)$. Hence verify that $\det(\Lambda^2 T) = (\det T)^2$.

Solution. By Definition 15.9.1 and Theorem 15.9.5, the entry of $[\Lambda^2 T]$ in row I and column J is $\det \mathbf{A}_{I,J}$, with rows and columns indexed by $\{1,2\}, \{1,3\}, \{2,3\}$ in that order. Computing the nine 2×2 minors,

$$[\Lambda^2 T] = \begin{pmatrix} 1 & 3 & 6 \\ -4 & 1 & 2 \\ -2 & -6 & 1 \end{pmatrix}.$$

Its determinant is 169. Expanding $\det \mathbf{A}$ along the first column (Theorem 7.6.1) gives $1 \cdot (1 - 0) + 2 \cdot (6 - 0) = 13$, and $13^2 = 169$.

Exercise. B2: Independence by minors

In $\mathbb{Q}^4$, let $\mathbf{v}_1 = (1,2,0,1)$, $\mathbf{v}_2 = (0,1,1,1)$, and $\mathbf{v}_3 = (1,0,-2,-1)$. Compute the coordinates of $\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \mathbf{v}_3$ in $\Lambda^3\mathbb{Q}^4$, and hence determine whether the list is linearly independent. Justify your answer.

Solution. Let $\mathbf{A} \in M_{4\times 3}(\mathbb{Q})$ have these columns. By Theorem 15.9.5 the four coordinates are the 3×3 minors on the row sets $\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}$. Computing them,

$$\det \mathbf{A}_{\{1,2,3\}} = \det \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \end{pmatrix} = 0,$$

and the same for the other three: $\det \mathbf{A}_{\{1,2,4\}} = \det \mathbf{A}_{\{1,3,4\}} = \det \mathbf{A}_{\{2,3,4\}} = 0$. So $\mathbf{v}_1 \wedge \mathbf{v}_2 \wedge \mathbf{v}_3 = 0$, and by Theorem 15.8.7 the list is **dependent**. Indeed $\mathbf{v}_3 = \mathbf{v}_1 - 2\mathbf{v}_2$.

Exercise. B3: Determinant of an inverse

Let $\dim V = n \geq 1$ and let $T \in \mathcal{L}(V)$ be invertible. Using Theorem 15.9.2 and Theorem 15.9.3, and no formula for the determinant, prove that $\Lambda^n T$ is invertible and that $\det(T^{-1}) = (\det T)^{-1}$.

Solution. By Theorem 15.9.2, $(\Lambda^n T)(\Lambda^n T^{-1}) = \Lambda^n(TT^{-1}) = \Lambda^n(\mathrm{id}_V) = \mathrm{id}$, and symmetrically in the other order. So $\Lambda^n T$ is invertible with inverse $\Lambda^n T^{-1}$. By Theorem 15.9.3 these two maps are multiplication by $\det T$ and by $\det(T^{-1})$ on the line $\Lambda^n V$, and their composite is multiplication by $\det T \cdot \det(T^{-1})$, which equals id. Since $\Lambda^n V \neq \{0\}$, this forces $\det T \cdot \det(T^{-1}) = 1$. In particular $\det T \neq 0$, and $\det(T^{-1}) = (\det T)^{-1}$.

15.9.7.3 C. Going deeper

Exercise. C1: The trace of an exterior power

Let $\mathbf{A} \in M_n(F)$ and $1 \leq k \leq n$, and let $T\colon F^n \to F^n$ be $\mathbf{x} \mapsto \mathbf{A}\mathbf{x}$.

(a) Prove that the matrix of $\Lambda^k T$ in the basis (e_I) has (I,J)-entry $\det \mathbf{A}_{I,J}$.

(b) Deduce that $\mathrm{tr}(\Lambda^k T)$ is the sum of the $\binom{n}{k}$ **principal** $k \times k$ minors of $\mathbf{A}$, those with $I = J$.

(c) Compute $\mathrm{tr}(\Lambda^2 T)$ for the matrix $\mathbf{A}$ of Example (The second exterior power of a 3×3 matrix), and check it against the matrix found there.

Solution.

(a) Column $J = \{j_1 < \cdots < j_k\}$ of the matrix is the coordinate vector of $\Lambda^k T(e_J) = \mathbf{a}_{j_1} \wedge \cdots \wedge \mathbf{a}_{j_k}$, where $\mathbf{a}_j = T\mathbf{e}_j$ is the j-th column of $\mathbf{A}$ (Definition 15.9.1). By Theorem 15.9.5 applied to the $n \times k$ matrix with columns $\mathbf{a}_{j_1}, \ldots, \mathbf{a}_{j_k}$, the coordinate at e_I is the determinant of its submatrix on rows I, which is $\det \mathbf{A}_{I,J}$ (Definition 7.7.1).

(b) The trace is the sum of the diagonal entries, that is, of the entries with $I = J$, and by (a) those are the principal minors $\det \mathbf{A}_{I,I}$.

(c) The three principal 2×2 minors of $\mathbf{A}$ are $\det\left(\begin{smallmatrix}2&1\\0&4\end{smallmatrix}\right) = 8$, $\det\left(\begin{smallmatrix}2&3\\5&1\end{smallmatrix}\right) = -13$, and $\det\left(\begin{smallmatrix}4&1\\2&1\end{smallmatrix}\right) = 2$. Their sum is -3, which is the sum of the diagonal entries $8, -13, 2$ of the matrix computed in Example (The second exterior power of a 3×3 matrix).

Exercise. C2: Cauchy–Binet from a wedge

Let $m \le n$, let $\mathbf{A} \in M_{m\times n}(F)$ and $\mathbf{B} \in M_{n\times m}(F)$, and let $T_{\mathbf{A}}\colon F^n \to F^m$ and $T_{\mathbf{B}}\colon F^m \to F^n$ be the associated maps.

(a) Show that $\Lambda^m T_{\mathbf{B}}(\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m) = \sum_J (\det \mathbf{B}_{J,[m]})\, e_J$, the sum over m-element subsets $J \subseteq [n]$.

(b) Show that $\Lambda^m T_{\mathbf{A}}(e_J) = (\det \mathbf{A}_{[m],J})\, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$ for each such J.

(c) Hence deduce the Cauchy–Binet formula $\det(\mathbf{AB}) = \sum_J \det \mathbf{A}_{[m],J} \det \mathbf{B}_{J,[m]}$ (Theorem 7.7.2).

Solution.

(a) $\Lambda^m T_{\mathbf{B}}(\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m) = \mathbf{Be}_1 \wedge \cdots \wedge \mathbf{Be}_m$ by Definition 15.9.1, and the columns of $\mathbf{B}$ are $\mathbf{Be}_1, \dots, \mathbf{Be}_m$. Apply Theorem 15.9.5 in F^n with $k = m$.

(b) Writing $J = \{j_1 < \cdots < j_m\}$, we get $\Lambda^m T_{\mathbf{A}}(e_J) = \mathbf{Ae}_{j_1} \wedge \cdots \wedge \mathbf{Ae}_{j_m}$, a wedge of m vectors of F^m. By Theorem 15.9.5 with $k = m = n$ there is only one index set, $[m]$ itself, and the coefficient is the determinant of the matrix with those columns, which is $\mathbf{A}_{[m],J}$.

(c) By Theorem 15.9.2, $\Lambda^m(T_{\mathbf{A}}T_{\mathbf{B}}) = (\Lambda^m T_{\mathbf{A}})(\Lambda^m T_{\mathbf{B}})$. The map $T_{\mathbf{A}}T_{\mathbf{B}}$ is the operator on F^m with matrix $\mathbf{AB}$, so by Theorem 15.9.3 the left side sends $\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$ to $\det(\mathbf{AB})\, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$. By (a) and then (b), the right side sends it to $\sum_J \det \mathbf{B}_{J,[m]} \det \mathbf{A}_{[m],J}\, \mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m$. Comparing coefficients on the basis vector $\mathbf{e}_1 \wedge \cdots \wedge \mathbf{e}_m \neq 0$ gives the formula.

Exercise. C3: Exterior powers of an injection

Let $T\colon V \to W$ be an **injective** linear map between finite-dimensional spaces and let $1 \le k \le \dim V$. Prove that $\Lambda^k T$ is injective. Is the analogous statement true with "injective" replaced by "surjective"?

Hint: choose a basis of V and use Theorem 15.8.5 in both spaces.

Solution. Let $(v_1, \dots, v_m)$ be a basis of V. Since T is injective, $(Tv_1, \dots, Tv_m)$ is linearly independent in W, so it extends to a basis $(Tv_1, \dots, Tv_m, w_{m+1}, \dots)$ of W (Theorem 1.6.5). By Theorem 15.8.5, the wedges $v_{i_1} \wedge \cdots \wedge v_{i_k}$ over increasing tuples form a basis of $\Lambda^k V$, and $\Lambda^k T$ sends each of them to $Tv_{i_1} \wedge \cdots \wedge Tv_{i_k}$, which is one of the basis vectors of $\Lambda^k W$ provided by the extended basis. So $\Lambda^k T$ carries a basis to a linearly independent list, hence is injective.

Yes for surjective as well, and more cheaply: if T is surjective then every $w_1 \wedge \cdots \wedge w_k$

equals $Tv_1 \wedge \cdots \wedge Tv_k$ for preimages v_i, so the image of $\Lambda^k T$ contains a spanning set of $\Lambda^k W$ (Theorem 15.8.5) and is therefore all of it.